Year 11 Higher

REVISION PACK

Shape and Space 5

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Shape and Space 5 · Circle theorems (Lessons 5.1, 5.2, 5.3)

Q	Working	Answer
the angle in a semicircle		
1.1	$\text{angle } ACB = 90^\circ$ $z = 180 - 90 - 31$ $z = 59^\circ$	$z = 59°$**. The angle in a semicircle is 90°.**
1.2	$\text{angle } ACB = 90^\circ$ $z = 180 - 90 - 42$ $z = 48^\circ$	$z = 48°$**. The angle in a semicircle is 90°.**
1.3	$\text{angle } ACB = 90^\circ$ $z = 180 - 90 - 23$ $z = 67^\circ$	$z = 67°$**. The angle in a semicircle is 90°.**
1.4	$\text{angle } ACB = 90^\circ$ $z = 180 - 90 - 36$ $z = 54^\circ$	$z = 54°$**. The angle in a semicircle is 90°.**
1.5	$\text{angle } ACB = 90^\circ$ $z = 180 - 90 - 19$ $z = 71^\circ$	$z = 71°$**. The angle in a semicircle is 90°.**
1.6	$\text{angle } ACB = 90^\circ$ $z = 180 - 90 - 48$ $z = 42^\circ$	$z = 42°$**. The angle in a semicircle is 90°.**
angles in the same segment		
1.7	$w = 46^\circ$	$w = 46°$**. Angles in the same segment are equal.**
1.8	$w = 29^\circ$	$w = 29°$**. Angles in the same segment are equal.**
1.9	$w = 53^\circ$	$w = 53°$**. Angles in the same segment are equal.**

1.10	$w = 35^\circ$	$w = 35°$**. Angles in the same segment are equal.**
1.11	$w = 41^\circ$	$w = 41°$**. Angles in the same segment are equal.**
1.12	$w = 62^\circ$	$w = 62°$**. Angles in the same segment are equal.**
cyclic quadrilaterals		
1.13	$x = 180 - 124$ $x = 56^\circ$	$x = 56°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
1.14	$x = 180 - 103$ $x = 77^\circ$	$x = 77°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
1.15	$x = 180 - 97$ $x = 83^\circ$	$x = 83°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
1.16	$x = 180 - 109$ $x = 71^\circ$	$x = 71°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
1.17	$x = 180 - 131$ $x = 49^\circ$	$x = 49°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
1.18	$x = 180 - 86$ $x = 94^\circ$	$x = 94°$**. Opposite angles of a cyclic quadrilateral add up to 180°.**
the alternate segment theorem		
1.19	$y = 68^\circ$	$y = 68°$**. Alternate segment theorem.**
1.20	$y = 57^\circ$	$y = 57°$**. Alternate segment theorem.**
1.21	$y = 74^\circ$	$y = 74°$**. Alternate segment theorem.**
1.22	$y = 63^\circ$	$y = 63°$**. Alternate segment theorem.**
1.23	$y = 49^\circ$	$y = 49°$**. Alternate segment theorem.**

1.24	$y = 78^\circ$	$y = 78°$**. Alternate segment theorem.**
the angle at the centre		
1.25	$150 \div 2 = 75^\circ$	**75°. The angle at the centre is twice the angle at the circumference.**
1.26	$96 \div 2 = 48^\circ$	**48°. The angle at the centre is twice the angle at the circumference.**
1.27	$124 \div 2 = 62^\circ$	**62°. The angle at the centre is twice the angle at the circumference.**
1.28	$108 \div 2 = 54^\circ$	**54°. The angle at the centre is twice the angle at the circumference.**
1.29	$136 \div 2 = 68^\circ$	**68°. The angle at the centre is twice the angle at the circumference.**
1.30	$84 \div 2 = 42^\circ$	**42°. The angle at the centre is twice the angle at the circumference.**